MILITARY SPECIFICATION SHEET
GENERATING SYSTEM, 25 KVA, 400 HERTZ, ALTERNATING CURRENT, INTEGRALLY EXCITED, BRUSHLESS

THIS SPECIFICATION SHEET IS APPROVED FOR USE BY THE NAVAL AIR SYSTEMS COMMAND, DEPARTMENT OF THE NAVY, AND IS AVAILABLE FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE.

THE COMPLETE REQUIREMENTS FOR PROCURING THE CABLE DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE LATEST ISSUE OF SPECIFICATION MIL-G-21480.

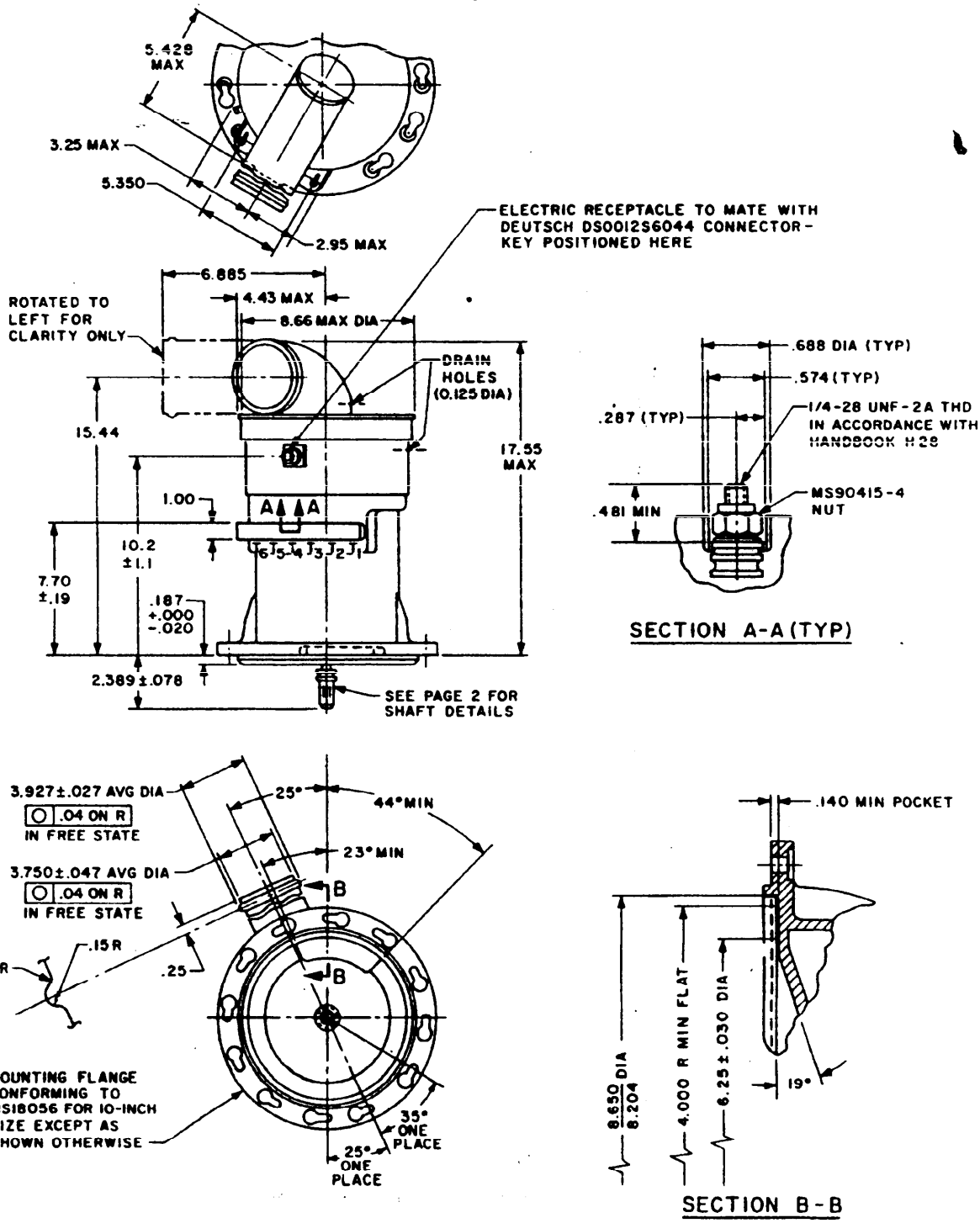
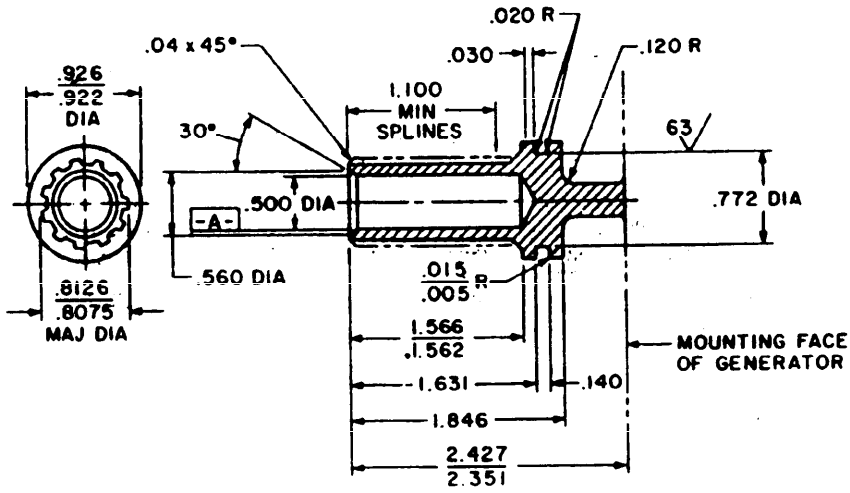


FIGURE 1. GENERATOR

P/N M21480/4-1

Source: https://assist.dla.mil - Downloaded: 2026-10-19T09:40Z
Check the source to verify that this is the current version before use.

FSC 6115



SPLINE DATA	
FORM	INVOLUTE
NUMBER OF SPLINES	12
PITCH	16/32 FLAT ROOT SIDE FIT
PRESSURE ANGLE	30°
PITCH DIA	.7500
MINOR DIA	.6710 +.0001 −.0050
MAJOR DIA	.8125 +.0001 −.0050

CIRCULAR SPLINE THICKNESS	
MAX EFFECTIVE	.0977
MIN ACTUAL	.0946
MAX TIF DIA	.6875
MIN MEAS OVER PINS	.9279
PIN DIA	.1200

NOTE: PITCH DIA OF SPLINES IS CONCENTRIC WITH -A- WITHIN .002 TIR.

GENERATOR SHAFT

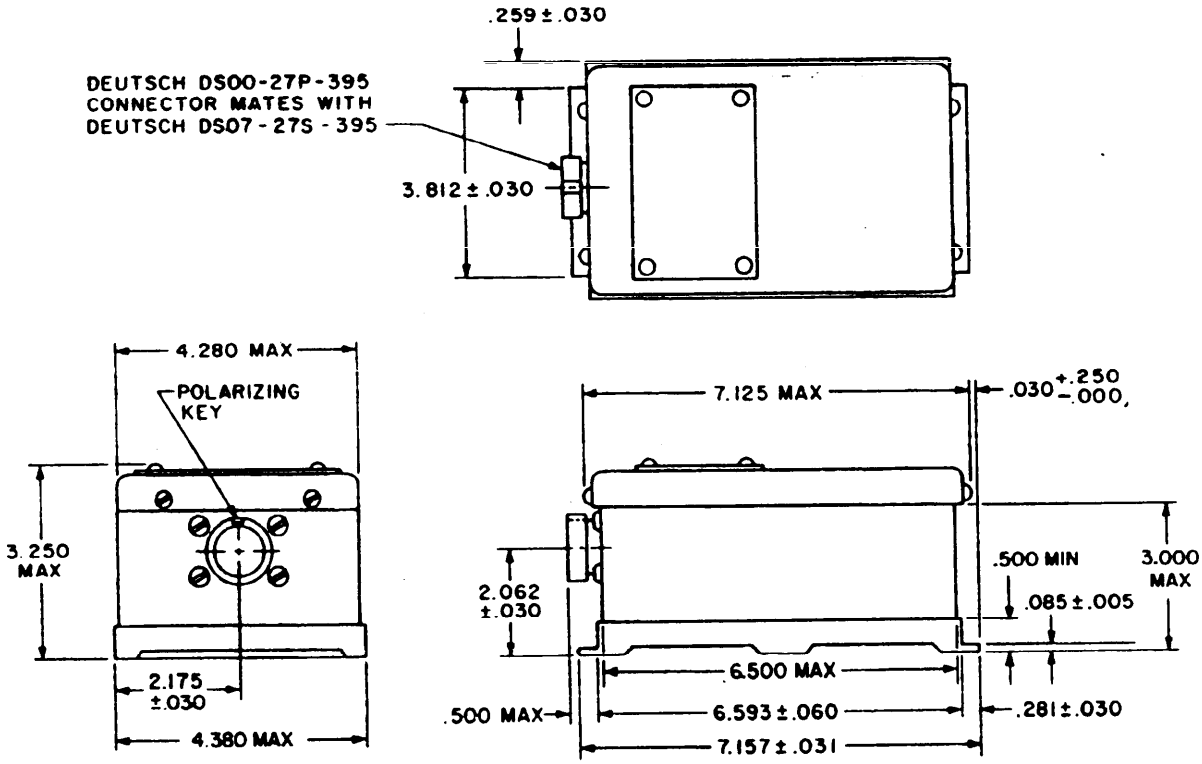


FIGURE 2. CONTROL UNIT - 3 LB MAX
P/N M21480/4-2

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:40Z
Check the source to verify that this is the current version before use.

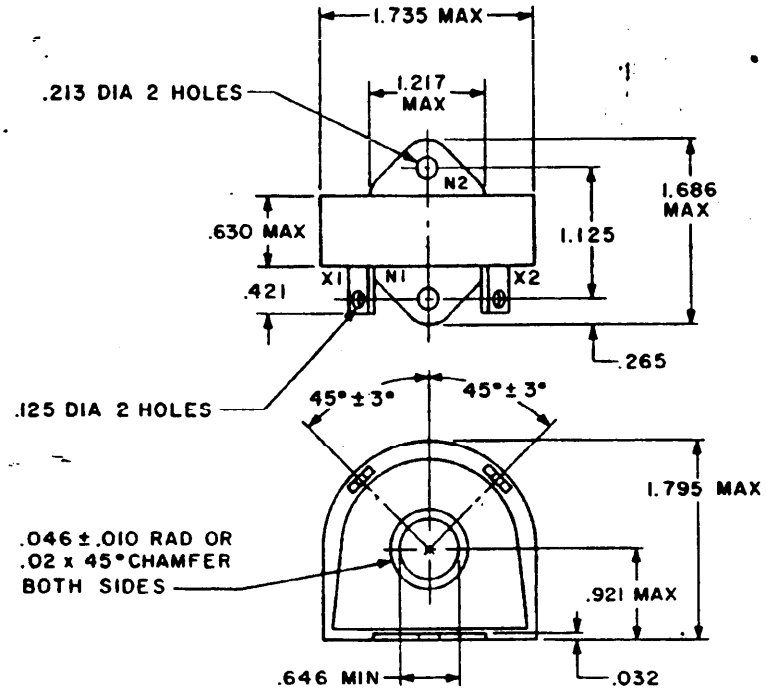
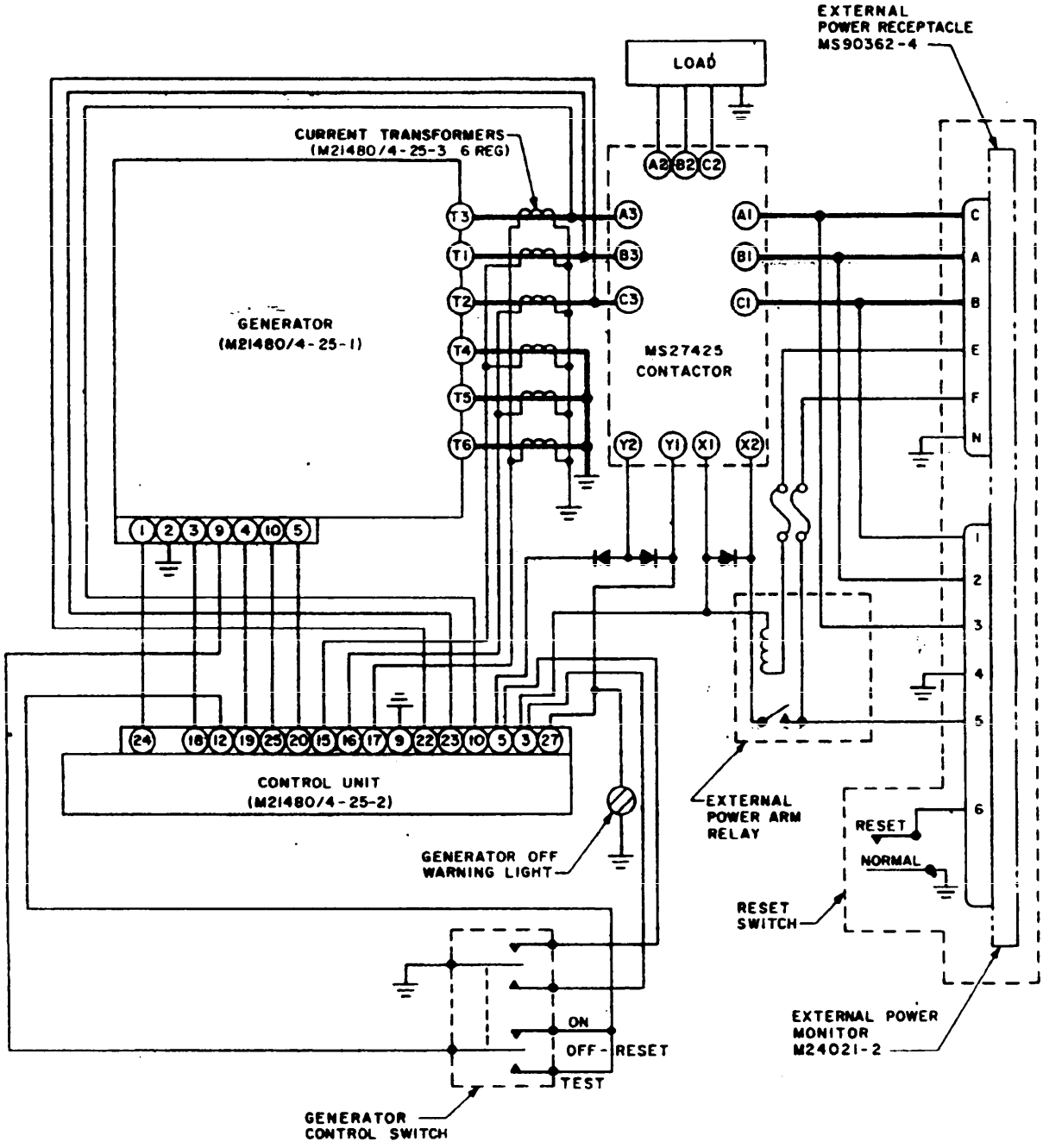


FIGURE 3. CURRENT TRANSFORMER - .125 LB MAX
P/N M21480/4-3

IDENTIFICATION OF ITEMS	
ITEM	MILITARY PART NUMBER
SYSTEM (CONSISTS OF ONE GENERATOR, ONE CONTROL UNIT, AND SIX CURRENT TRANSFORMERS)	M21480/4-25
GENERATOR	M21480/4-25-1
CONTROL UNIT	M21480/4-25-2
CURRENT TRANSFORMER	M21480/4-25-3

GENERATOR DATA	
RATED VOLTAGE	120/208
RATED LOAD	25 KVA
PHASE	3
FREQUENCY (Hz)	380-420
SPEED RANGE (RPM)	7600-8400
MAX SPEED FOR REGULATION (RPM)	9000
RATED POWER FACTOR	0.75 LAG-1
MAX WEIGHT, POUNDS	56.5
MAX OVERHUNG MOMENT	350 IN-LBS
OVERSPEED (RPM)	10,000

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:40Z
Check the source to verify that this is the current version before use.



NOTE : COMPONENTS SHOWN IN DOTTED AREAS
ARE NOT PART OF THIS SYSTEM

FIGURE 4. TEST CIRCUIT

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:40Z
Check the source to verify that this is the current version before use.

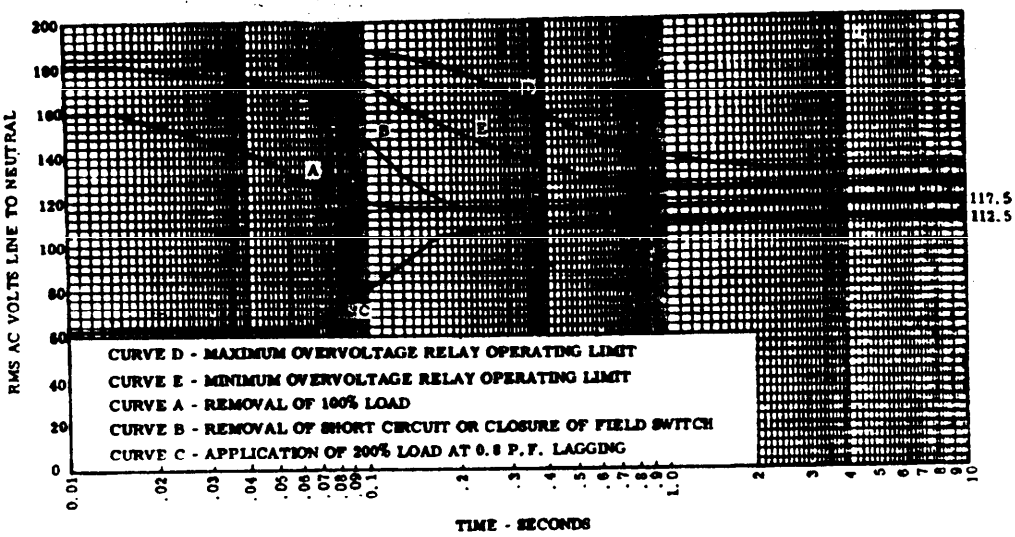


FIGURE 5. System voltage limits

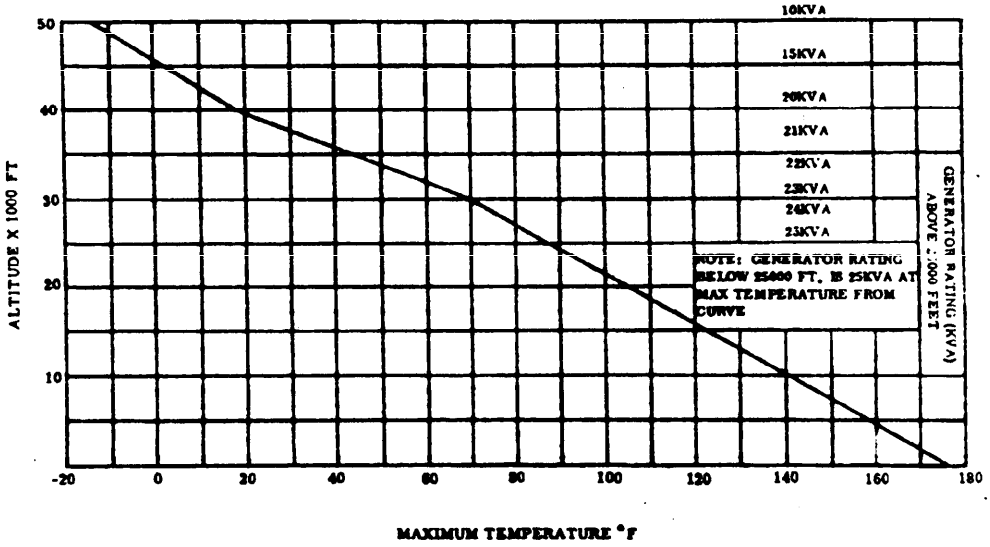
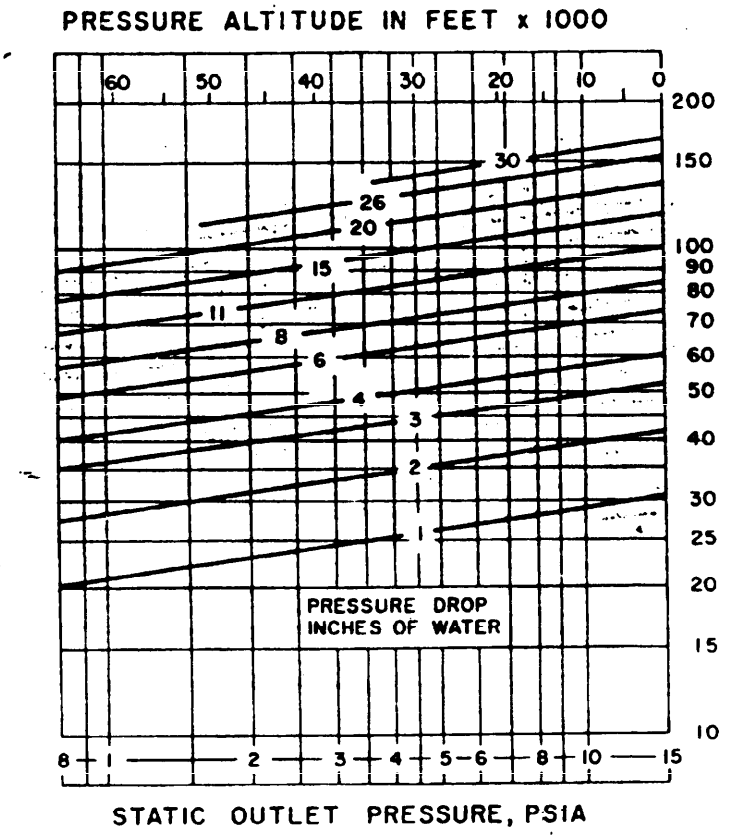


FIGURE 6. Generator output (KVA) vs altitude and temperature for self-cooled operation

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:40Z
Check the source to verify that this is the current version before use.



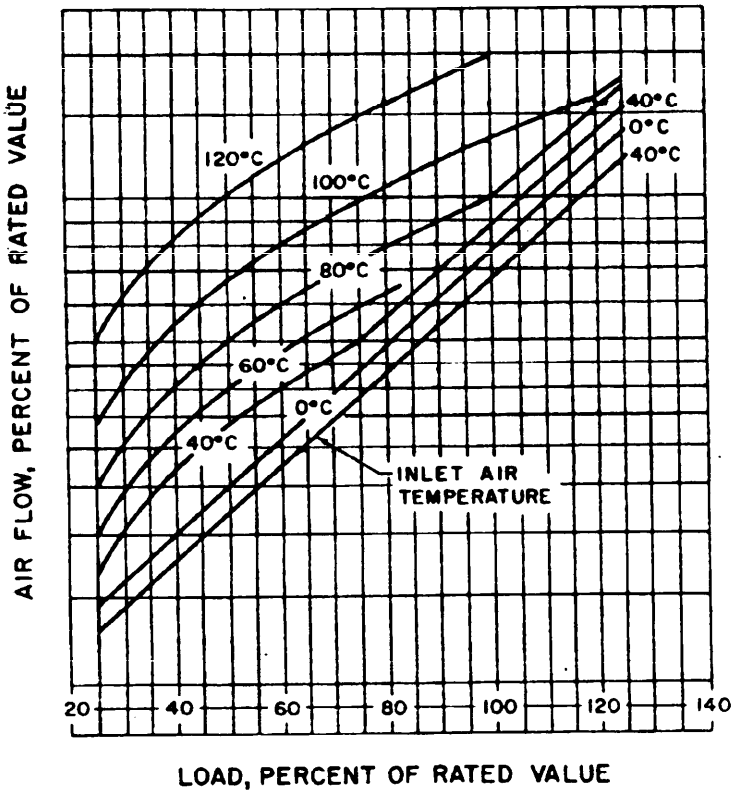


FIGURE 7. GENERATOR RATING CHART

Source: https://assist.dla.mil -- Downloaded: 20260-19T09:40Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

1. The requirements of MIL-G-21480 apply with the following exceptions to existing paragraphs:

Delete the following paragraphs:

3.4.2.1, 3.4.2.2, 3.4.2.3, 3.4.2.4, 3.4.2.5, 3.4.2.6, 3.4.2.7, 3.4.2.8, 3.4.10.5, 3.4.10.6, 3.4.15, 3.5.7.5, 3.5.9, 4.5.6, 4.5.19, 4.5.20, 4.5.22.1, 4.5.22.2, 4.5.22.3, 4.5.22.3.1, 4.5.22.3.2, 4.5.22.3.3, 4.5.22.3.4, 4.5.23.1, 4.5.23.1.1, 4.5.23.1.2, 4.5.23.2, 4.5.24.1, 4.5.24.2, 4.5.24.2.1, 4.5.24.2.2, 4.5.24.2.3, 4.5.24.2.4, 4.5.24.2.5, 4.5.24.3, 4.5.24.3.1, 4.5.24.3.2, 4.5.24.3.3, 4.5.24.3.4, 4.5.25.1, 4.5.25.2, 4.5.25.2.1, 4.5.26, 4.5.28, 4.5.29, 4.5.32, 4.5.33, 4.5.33.1, 4.5.33.2.

3.4.2 Delete and add: The system shall meet the requirements of MIL-E-81910 with the following exceptions. For all conditions the lowest temperature shall be -55°C; for the generator, the limits of Figure 6 of the specification sheet shall be used. For generator vibration testing, the generator will be energized at rated speed no load and the amplitude of vibration during the main bending mode shall be limited to 20g's monitored at the anti-drive end of the generator but the input vibration shall not be lowered below 5g's.

3.4.8 Delete first sentence and add: The generator shall be three phase, wye-connected, 400 cps, 120/208 volt, air-cooled, brushless, and shall be compatible with a constant speed drive conforming to Military Part Number M7101/1-60.

3.4.8.1 Delete and add: Self-Cooled. The generator shall be capable of continuous output in accordance with Figure 6 of the specification sheet. Ram Air Cooled. The generator shall be capable of continuous 25 KVA output from sea level to 50,000 feet when supplied with ram air in accordance with Figure 7 of the Specification Sheet.

3.4.8.2 Delete first sentence and add: The system shall be capable of delivering 150 percent rated current at any power factor from 0.75 lagging to unity, for 2 minutes.

3.4.9.1 Add: The mounting flange shall be as specified in Figure 1 of the specification sheet. A package holding one MS90415-6 nut for each mounting hole in the generator flange shall be securely attached to each generator in such a way that it need not be removed for checkout of the component prior to installation on the aircraft. The coupling spline shall be as specified in Figure 1 of the specification sheet.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:40Z
Check the source to verify that this is the current version before use.

In separate nonconforming area exists and is clearly marked

3.4.9.2 Delete and add: A shear section shall be incorporated between the drive spline of the generator and the rotor. The shear section shall fail at a torque value of 1100 ± 150 inch-pounds. Flexible drive provisions are not required.

3.4.9.3 Add: Electric connections shall be in accordance with Figures 1 and 4 of the specification sheet.

3.4.9.4 Delete and add: The direction of rotation shall be clockwise when viewing the generator drive end. See Figure 1 of the specification sheet.

3.4.9.6.1.1 Change "10,000 hours" to "1,000 hours".

3.4.9.6.1.2 Self-Cooling. An integral fan shall be provided to provide self-cooling during all conditions from static idle (sea level) to 50,000 feet.

3.4.10 Delete and add: Accessory components are defined as generator control equipment consisting of a GCU and six current transformers. The GCU shall consist of a supervisory panel, voltage regulator, and differential protection unit. The regulator shall sense and regulate average phase voltage.

3.4.10.1 Add: The regulator voltage setting shall be preset at 117 ± 1.0 volt at 25 ± 5°C, 400 Hz, full rated load, P.F. = 1.0.

3.4.10.4 Delete and add: The GCU and current transformers shall meet the requirements of the specification sheet when mounted in any position. Mounting provisions shall be in accordance with Figures 2 and 3 of the specification sheet.

3.4.11 Change "All system components" to "Generator and GCU".

3.4.13 The nomenclature shall be as follows:

Generator

A.C. Generator 25 KVA
Volts: 120/208
P.F.: 0.75 to 1.0
Phase: 3
Frequency: 380/420 Hz
Manufacturer's Part Number
Manufacturer's Serial Number
Military Part Number: M21480/4-25-1
Contract Number
Date of Manufacture
Manufacturer's Name or Trademark
Federal Stock Number
US

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:40Z
Check the source to verify that this is the current version before use.

Control Unit

115/200 volts AC
3 Phase
400 Hz
Manufacturer's Part Number
Manufacturer's Serial Number
Military Part Number: M21480/4-25-2
Contract Number
Date of Manufacture
Manufacture's Name or Trademark
Federal Stock Number
US

Current Transformer

Manufacturer's Part Number
Manufacturer's Serial Number
Military Part Number: M21480/4-25-3
Contract Number
Date of Manufacture
Manufacturer's Name or Trademark
Federal Stock Number
US

3.4.17 Add: Reliability. The equipment shall be designed for, and capable of demonstration of, a Mean-Time-Between-Failures Reliability Index of 5,000 hours for the generator, 2,700 hours for the generator control unit and 16,700 hours for the current transformers, based on applicable flight data, while operating in an environment at least as severe as that expected under operational usage conditions. This requirement shall be capable of achievement at any time during the required operating life and after exposure to the specified storage conditions. Further, the equipment reliability shall not be degraded below the required level as a result of normal field usage and maintenance. Reliability analyses, in accordance with MIL-STD-756, shall be submitted within 120 days of contract award and updated periodically thereafter. These analyses shall indicate the degree of conformance of the equipment design to the specified reliability level.

3.5.1 Delete and add: Voltage limits shall be as shown on Figure 5 of the specification sheet.

3.5.4 Delete and add: The phase voltage wave form shall be within the following limits for all normal operating conditions of the system.

(1) Crest Factor 1.41 $\pm$ 0.1
(2) Total rms harmonic content 4% of fundamental (rms)
(3) Any single harmonic not to exceed 3% of the fundamental rms

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:40Z
Check the source to verify that this is the current version before use.

3.5.5 Delete last sentence and add: Ampere requirements for external use of the line contractor and a generator indicator shall be a minimum of 2 amperes for 1 second (maximum) and a continuous current of 0.5 amperes.

3.5.7.2 Delete all after "defined by" and add: Figure 5, Curve D of the specification sheet.

3.5.7.3 Delete and add: The system shall trip when any one phase voltage drops to a value of 95-105 volts for a time delay of 3-7 seconds. The system shall be capable of being reset when all three phase voltages are at least 1.5 volts above drop-out level and before any phase exceeds 110 volts.

3.5.7.4 Delete and add: The system shall provide a means of preventing a frequency of less than 370 $\pm$ 5 Hz being impressed on the load bus for a period of 2.5 $\pm$ 0.5 seconds. Pickup value shall be at least 1 Hz greater than drop out level and before the frequency exceeds 385 Hz.

3.5.7.6 Change "exceeds 25 amperes" to "is 40 $\pm$ 4 amperes".

Add: 3.9 Semi-conductor operating temperature. Semi-conductors used in the system shall be selected such that operating temperatures under all conditions specified herein shall be below the maximum allowable operating temperature, based on a design life of 2500 hours.

Add: 3.10 Store life. It shall be a design objective that the generator and generator control panel meet the requirements of this specification after being stored for a period of not less than 5 years in a world-wide environment as defined in MIL-STD-210.

4.4.1.3.2 Delete and add: The ambient air temperature and temperature of all surrounding objects shall be maintained at 85 $\pm$ 5°C. The ambient air velocity shall be less than 5 knots actual airspeed.

4.4.3.4 Change "5 volt drop" to "2.5 volt drop".

4.5.3 Change "Figure 3" to "Figure 5 of the specification sheet".

4.5.9 Delete all after (d) and add:

(d) At minimum, maximum and rated generator RPM.

The efficiency at rated load and speed shall not be less than 80%. The efficiency shall not be less than 77% under any condition greater than 1/2 rated current. The efficiency at 1/4 rated current shall not be less than 70%.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:40Z
Check the source to verify that this is the current version before use.

4.5.17 Add: The generator control equipment shall be subejcted to a life cycling test at 81°C and at -55°C. The equipment shall be subjected to 5,000 cycles of operation at each temperature. Tests shall be conducted at the maximum rating of the equipment.

4.5.20.1 Delete and add: The CU and CT shall be subjected to a Salt Fog test in accordance with MIL-E-81910, paragraph 4.6. At the completion of this test the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200% load, and protective functions.

4.5.20.2 Delete and add: The generator shall be subjected to an Oil-Salt Water Ingestion test in accordance with MIL-E-81910, paragraph 4.6.1. At the completion of this test the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200% load, and protective functions.

4.5.21 Delete and add: The generator, CU, and CT shall be subjected to a Fungus Test in accordance with MIL-E-81910, paragraph 4.3 with the installation instructions and mounting nuts (as required by 4.3.1) attached. At the completion of this test the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200% load and protective functions. Mounting nuts are to be unaffected and the package holding the nuts suitable for shipment. Instructions are to be legible.

4.5.22 Delete and add: The generator shall be subjected to a Dust Test in accordance with MIL-E-81910, paragraph 4.2. At the completion of this test the unit shall successfully demonstrate its buildup characteristics, the effects of application and removal of rated and 200% load, and protective functions.

4.5.23 Add: The generator shall be subjected to a Vibration Test in accordance with MIL-E-81910, paragraph 4.9 except that the amplitude of vibration monitored at the anti-drive end of the generator during the main bending mode shall be limited to 20g's but the vibration input shall not be lowered below 5g's. At the completion of this test the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200% load, and protective functions.

4.5.24 Delete and add: The generator, CU, and CT shall be subjected to a Shock Test in accordance with MIL-E-81910, paragraph 4.7. At the completion of Procedure I, the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200% load, and protective functions. There shall be no failure of the mounting attachments, and the test item shall remain in place and not create hazard during Procedure III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:40Z
Check the source to verify that this is the current version before use.

4.5.25 Delete and add: The generator, CU, and CT shall be subjected to a humidity test in accordance with MIL-E-81910, paragraph 4.4. At the completion of this test the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200% load, and protective functions.

4.5.27 Add: Failure shall occur within the torque limits specified in paragraph 3.4.9.2 of the specification sheet.

4.5.30 Delete and add: The generator, CU and CT shall be subjected to an acceleration test in accordance with MIL-E-81910, paragraph 4.1. At the completion of this test the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200% load, and protective functions.

2. The color shall conform to FED-STD-595 color 17875.

NOTES:

1. Dimensions in inches. Unless otherwise specified, tolerances: decimals $\pm$ .06, angles $\pm$ 1°.

Preparing activity:
Navy - AS
(Project No. 6115-N463)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:40Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12063 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Naval Air Engineering Center
ESSD (Code 9313)
Lakehurst, NJ 08733

FOLD

☆ U.S. GOVERNMENT PRINTING OFFICE: 1981—705-063/4283 2-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:40Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-G-21480/4C

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:40Z
Check the source to verify that this is the current version before use.